1.
On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

Ans:

Step 1: Open Cisco Packet Tracer, place one 2960 Switch and three PCs (PC-Insta, PC-Zomato, and PC-Flipkart) onto the workspace.

Step 2: Click on the Switch, navigate to the CLI tab, and create the three required VLANs by typing the following commands:

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name Insta

Switch(config-vlan)# vlan 20

Switch(config-vlan)# name Zomato

Switch(config-vlan)# vlan 30

Switch(config-vlan)# name Flipkart

Switch(config-vlan)# exit

Step 3: Bind the switch port FastEthernet0/1 (connected to PC-Insta) to VLAN 10:

Switch(config)# interface FastEthernet0/1

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 10

Switch(config-if)# exit

Step 4: Bind the switch port FastEthernet0/2 (connected to PC-Zomato) to VLAN 20:

Switch(config)# interface FastEthernet0/2

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 20

Switch(config-if)# exit

Step 5: Bind the switch port FastEthernet0/3 (connected to PC-Flipkart) to VLAN 30:

Switch(config)# interface FastEthernet0/3

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 30

Switch(config-if)# exit

Step 6: Assign IP addresses to the PCs matching their subnets: Configure PC-Insta with 192.168.10.10 (Gateway: 192.168.10.1), PC-Zomato with 192.168.20.10 (Gateway: 192.168.20.1), and PC-Flipkart with 192.168.30.10 (Gateway: 192.168.30.1).

2.

Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

Ans:

 Step 1: Add a Router to the workspace and connect its GigabitEthernet0/0 interface to the switch's GigabitEthernet0/1 interface.

 Step 2: Click on the Router, open the CLI tab, turn on the main physical port, and leave it without a base IP address:

Router# configure terminal

Router(config)# interface GigabitEthernet0/0

Router(config-if)# no shutdown

Router(config-if)# exit

Step 3: Configure the sub-interface for VLAN 10 (Insta) by linking it to 802.1Q encapsulation and setting the default gateway IP:

Router(config)# interface GigabitEthernet0/0.10

Router(config-subif)# encapsulation dot1Q 10

Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config-subif)# exit

Step 4: Configure the sub-interface for VLAN 20 (Zomato) using the same structure:

Router(config)# interface GigabitEthernet0/0.20

Router(config-subif)# encapsulation dot1Q 20

Router(config-subif)# ip address 192.168.20.1 255.255.255.0

Router(config-subif)# exit

Step 5: Configure the sub-interface for VLAN 30 (Flipkart) using the same structure:

Router(config)# interface GigabitEthernet0/0.30

Router(config-subif)# encapsulation dot1Q 30

Router(config-subif)# ip address 192.168.30.1 255.255.255.0

Router(config-subif)# exit

3.

Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command.

Hint: Use the switchport mode trunk command on the relevant interface.

Ans:

 Step 1: Return to the Switch CLI configuration terminal window.

 Step 2: Access the specific physical port interface connected directly to the router (GigabitEthernet0/1).

 Step 3: Turn on trunking parameters on that port to allow all VLAN tag traffic to travel over the single cable line:

Switch(config)# interface GigabitEthernet0/1

Switch(config-if)# switchport mode trunk

Switch(config-if)# exit

 Step 4: Return to privileged EXEC mode by typing end.

 Step 5: Run the verification status command to verify the trunk link configuration details:

Switch# show interfaces trunk

Step 6: Look at the screen output text to verify that interface Gi0/1 is successfully listed as "status: trunking".

4.

Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

Ana:

 Step 1: Click on PC-Insta (192.168.10.10), navigate to the Desktop tab, and open the **Command Prompt**.

 Step 2: Type ping 192.168.20.10 to send test data packets to the PC located in the Zomato VLAN.

 Step 3: Observe that the ping **succeeds** completely after a brief initial ARP delay window.

Even though the PCs are on isolated logical networks, the ping works because data packets leave PC-Insta and go up to the Router's Gi0/0.10 gateway sub-interface. The router reads the destination network address, references its active routing table, switches the traffic internally, and forwards it down through the trunk link via sub-interface Gi0/0.20 to reach PC-Zomato.

5.

Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

Ans:

 Step 1: To avoid disrupting the existing network traffic, do not shut down or change any of the currently configured interfaces.

 Step 2: Open the Switch CLI terminal window and add the new VLAN entry profile definitions:

Switch# configure terminal

Switch(config)# vlan 40

Switch(config-vlan)# name Spotify

Switch(config-vlan)# exit

Step 3: Bind a vacant switch port (e.g., FastEthernet0/4) to the new Spotify VLAN 40:

Switch(config)# interface FastEthernet0/4

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 40

Switch(config-if)# exit

Step 4: Open the Router CLI window and build a completely new independent sub-interface for the Spotify network link path:

Router(config)# interface GigabitEthernet0/0.40

Router(config-subif)# encapsulation dot1Q 40

Router(config-subif)# ip address 192.168.40.1 255.255.255.0

Router(config-subif)# exit

Step 5: Connect a fourth PC (PC-Spotify) to Switch port Fa0/4 and assign it IP 192.168.40.10 (Gateway: 192.168.40.1). This expands the network seamlessly without any service downtime.